

Linearization

$\dot{x} = Ax + Bu$. Current system is non-linear

$$\dot{x} = f(x, u).$$

Assuming theta is small: $\sin \theta \approx \theta$, $\cos \theta \approx 1$

Discarding θ^2 , $\dot{\theta}^2 \theta$ etc.

$$\dot{p} = \frac{F - mg \sin \theta \cos \theta + mL \dot{\theta}^2 \sin \theta}{M + m - m \cos^2 \theta}$$

$$\dot{p} = \frac{F - mg\theta}{M} \Rightarrow \ddot{\theta} = \frac{g}{L} \theta - \frac{F - mg\theta}{mL}$$

$$\dot{p} = -\frac{mg}{M} \theta + \frac{1}{m} F$$

$$\ddot{\theta} = \frac{g}{L} \theta + \frac{mg}{mL} \theta - \frac{F}{mL}$$

$$\ddot{\theta} = \frac{(M+m)g}{mL} \theta - \frac{1}{mL} F$$

$$\dot{x} = Ax + Bu, \quad x = \begin{bmatrix} p \\ \dot{p} \\ \theta \\ \dot{\theta} \end{bmatrix}, \quad \dot{x} = \begin{bmatrix} \dot{p} \\ \ddot{p} \\ \dot{\theta} \\ \ddot{\theta} \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{mg}{m} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{M+m}{mL}g & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ \frac{1}{m} \\ 0 \\ -\frac{1}{mL} \end{bmatrix}$$